

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CE) Examination

March-April 2018

CE409: Wireless Communication & Mobile Computing**Date: 30/03/2018, Friday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make & specify suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Do as Directed. [05]**

- (a) _____ is the common protocol for Internet. [01]
- (b) What is Signal? What is the role of it in communication system to exchange data? [02]
- (c) Why can waves with a very low frequency follow the earth's surface? [02]

Q - 2 Do as Directed (Any Three). [15]

- (a) What is Ad-hoc and Infrastructure mode in WLAN? Explain them in detail.
- (b) What are applications of GSM? List and explain such five applications.
- (c) What is The DHCP Process for Client Server?
- (d) What is Agent advertisement in Mobile IP? Explain it in detail.
- (e) Compare and contrast 3G and GSM.

Q - 3 Do as Directed (Any Three). [15]

- (a) Explain any two mechanisms of the transmission control protocol (TCP) that influence the efficiency of TCP in a mobile environment.
- (b) Explain working of Tunneling mobile IP.
- (c) Give any five characteristics of Mobile Ad Hoc Network.
- (d) Write a short note on MAC address Access Control in WLAN.
- (e) What are the applications of Mobile Computing? Explain them in detail.

SECTION – II**Q - 4 Do as Directed. [05]**

- (a) One of the most commonly used MAC scheme for wired networks is _____ [01]
- (b) What are the main problems when transmitting data using wireless system that were made for voice transmission? [02]
[02]
- (c) Compare TCP and UDP.

Q - 5 Do as Directed (Any Three).

[15]

- (a) Write down any five Environmental Challenges in Mobile Communication.
- (b) What are the reasons for choosing WAP (Wireless Application Protocol)?
- (c) State and compare any two transmission loss in TCP that can occur in wired and wireless networks.
- (d) Explain the advantages of I-TCP (indirect transmission protocol) in detail.
- (e) Draw and explain IEEE 802.11 MAC packet structure.

Q - 6 Do as Directed (Any Three).

[15]

- (a) Draw and explain GPRS architecture reference model.
- (b) Draw and explain Bluetooth Protocol Stack.
- (c) Draw and explain UMTS system architecture.
- (d) List down task of following protocols in GPRS:
GTP, SNDTCP, BSSGP, RLC, PDTCHs
- (e) What are SGSN and GGSN in GRPS? Explain them in detail.
